UNIT - I

Energy Theorems and Deflections

1. Introduction

- Structural analysis finds **forces (axial, shear, bending moment, torsion)** and **deformations (deflection, slope)** in a structure under loads.
- There are two broad approaches: **Force (flexibility) methods** and **Displacement (stiffness) methods** . Energy methods give a powerful tool for both.
- **Energy methods** use the principle: *work done by external loads = strain energy stored in the structure* (for a conservative, linear elastic system).
- Where equilibrium equations alone give deflection only with difficulty (trusses, frames, curved beams), energy theorems give a quick and systematic answer.

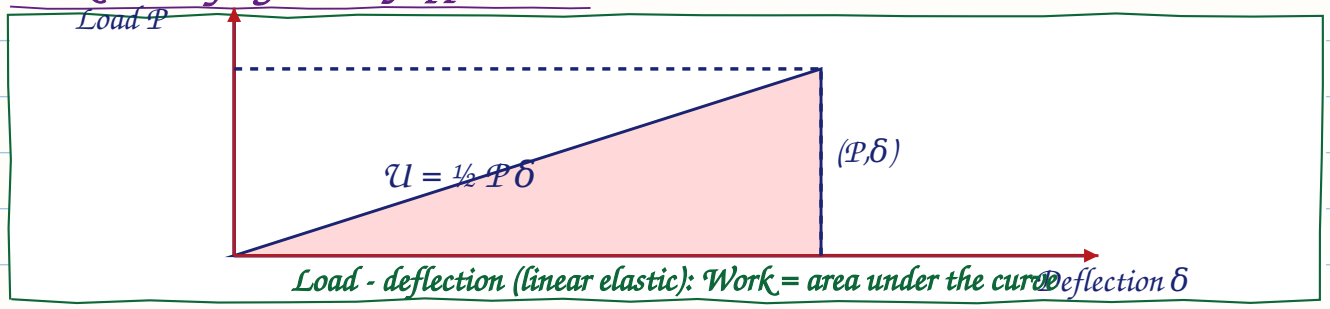
Basic assumptions

1. Material is **linearly elastic** (Hooke's law valid) 2. Deflections are **small** 3. Supports are unyielding (unless stated) 4. Principle of **superposition** holds 5. Loads are applied **gradually** (no dynamic effect).

2. Work, Energy and Strain Energy

- **Work** = force × displacement in the direction of the force.
- **Strain energy (U)** is the energy stored in a deformed elastic body due to the work done by external loads. It is recovered completely when the load is removed (elastic).

Work done by a gradually applied load



- Work done = area of the triangle = **½ P δ** . This is the strain energy stored.
- For a suddenly applied load, work = P δ — twice as much, which is why the sudden-load stress is double.

Important terms

Resilience = total strain energy stored in the body. **Proof resilience** = max strain energy stored at the elastic limit. **Modulus of resilience** = proof resilience per unit volume = $\sigma^2 / 2E$.

3. Strain Energy in a Linear Elastic System

- **Clapeyron's theorem:** For a linear elastic structure under several loads $P_1, P_2, \ldots$ producing deflections $\delta_1, \delta_2, \ldots$ in the direction of the loads (all applied gradually together):

Clapeyron's theorem of work

$U = \frac{1}{2}(P_1\delta_1 + P_2\delta_2 + \ldots) = \frac{1}{2}\Sigma P\delta$

- Strain energy is always **positive** and is a **quadratic function** of load ($U \propto P^2$). So **superposition does not apply to strain energy** — you cannot add U of two load cases separately.
- Strain energy per unit volume under direct stress: $u = \sigma^2/2E = \frac{1}{2}\sigma\varepsilon$.

4. Expressions for Strain Energy

(a) Due to axial load

- For a bar of length L, area A, modulus E carrying axial force P:

Axial load

$U = P^2 L/2AE$ (uniform bar)

$U = \int P^2\,dx/2AE$ (when P or A varies along length)

$U = \Sigma P_i^2 L_i/2A_iE_i$ (bar of several segments)

- Derivation: Stress $\sigma = P/A$, strain $\varepsilon = P/AE$; $u = \sigma^2/2E$; $U = u \times$ volume $= (P^2/2A^2E) \times AL = P^2L/2AE$.

(b) Due to bending moment

- Consider an element dx of a beam with bending moment M. Rotation between the two faces $d\theta = M\,dx/EI$. Work $= \frac{1}{2}M\,d\theta$.

Bending

$U = \int M^2\,dx/2EI$ (integrate over the full length of the member)

- Derivation: Since: $dU = \frac{1}{2}M\,d\theta = \frac{1}{2}M \times (M\,dx/EI) = M^2\,dx/2EI$.

(c) Due to shear force

- Average shear stress $\tau = V/A$; strain energy per unit volume $= \tau^2/2G$. A form factor α accounts for non-uniform shear stress across the section.

Shear

$U = \int \alpha V^2\,dx/2GA$ where α = form factor for shear

- $\alpha = 6/5 = 1.2$ for rectangle, $10/9$ for solid circle, about **1.0 for I-section** (area of web only), **2.0** for thin tube.

(d) Due to torsion (for completeness)

Torsion

$U = \int T^2\,dx / 2GJ$ (for a shaft: $U = T^2\,L / 2GJ$)

Total strain energy in a member

Total

$U = \int N^2\,dx / 2AE + \int M^2\,dx / 2EI + \int \alpha V^2\,dx / 2GA + \int T^2\,dx / 2GJ$

→ In beams, the shear and axial energies are very small compared to bending energy, so only $U = \int M^2\,dx / 2EI$ is used. In trusses, only axial energy exists. In frames, axial + bending (shear neglected).

Type of member	Energy to consider	Expression
Beam	bending (shear neglected)	$\int M^2\,dx / 2EI$
Pin-jointed truss	axial only	$\Sigma F^2\,L / 2AE$
Rigid-jointed frame	bending (+ axial if needed)	$\int M^2\,dx / 2EI$
Curved bar / ring	bending + axial + shear	all three terms
Shaft	torsion	$T^2\,L / 2GJ$

5. Strain Energy due to Impact and Sudden Load (extra points)

- **Gradual load P:** stress $\sigma = P/A$; $U = \tfrac{1}{2} P \delta$.
- **Sudden load:** stress = $2P/A$ (double); extension = $2P\,L / AE$.
- **Impact load** (weight W falling through height h on a bar of length L):

Instantaneous stress

$\sigma = (W/A)[\,1 + \sqrt{(\,1 + 2\,h\,A\,E / W\,L\,)}\,]$; if $h = 0$ then $\sigma = 2W/A$.

6. Castigliano's Theorems

- Proposed by **Carlo Alberto Castigliano** (Italian engineer, 1873).

Castigliano's first theorem (deflection form, as in most Indian syllabi)

"The partial derivative of the total strain energy of a structure, with respect to any one of the applied loads, gives the displacement of the point of application of that load, in the direction of that load."

$\delta = \partial U / \partial P$ and for a moment: $\theta = \partial U / \partial M$

- Here U must be written in terms of the loads (as a function of P). P can be a force (gives linear deflection) or a couple (gives rotation).

Note on naming

Some books call this the **Castigliano's second theorem** ($\delta = \partial U/\partial P$) and call the "force = $\partial U/\partial \delta$" form the first theorem. In our syllabus: **Castigliano's I = deflection theorem ($\delta = \partial U/\partial P$)** and **Castigliano's II = theorem of least work ($\partial U/\partial R = 0$)**, used in Unit II.

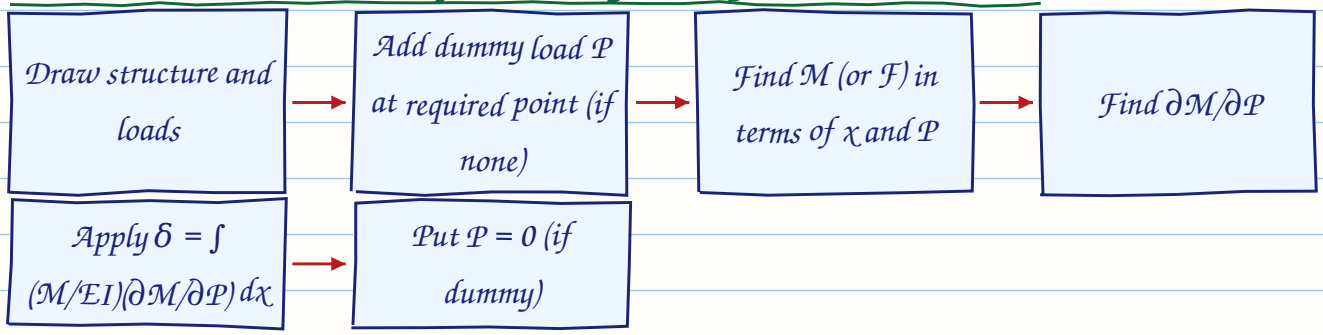
Proof (short)

- Loads $P_1 \dots P_n$ act and the strain energy is U. Let P_n increase by dP_n: the strain energy increases by $dU = (\partial U/\partial P_n)\, dP_n$.
- Now apply dP_n **first** and then the original loads: the energy is $\tfrac{1}{2}\, dP_n\, d\delta_n + \delta_n\, dP_n + U$. Energy stored does not depend on the order of loading, so equate both and neglect the second-order term $\tfrac{1}{2}\, dP_n\, d\delta_n$.
- Hence $\partial U/\partial P_n = \delta_n$.

Conditions for use

- Elastic, linear system; supports do not move; temperature constant.
- The deflection is to be found **at the point and in the direction of an existing load**. If there is no load at that point → **apply a dummy (fictitious) load P** there, find $\partial U/\partial P$, then put $P = 0$.

7. Procedure to Find Deflection by Castigliano's Theorem



1. Write the bending moment M in each segment (with limits of x) including P.
2. **Differentiate first, integrate later** — it avoids squaring: $\partial U/\partial P = \partial/\partial P \int M^2 dx/2EI = \int (M/EI)(\partial M/\partial P)\, dx$.
3. Integrate segment by segment and add.
4. Substitute $P = 0$ for the dummy load. A **positive** answer means deflection is in the direction of P.

- For a rotation: apply a dummy moment M_0 at the point; $\theta = \int (M/EI)(\partial M/\partial M_0)\, dx$ with $M_0 = 0$.

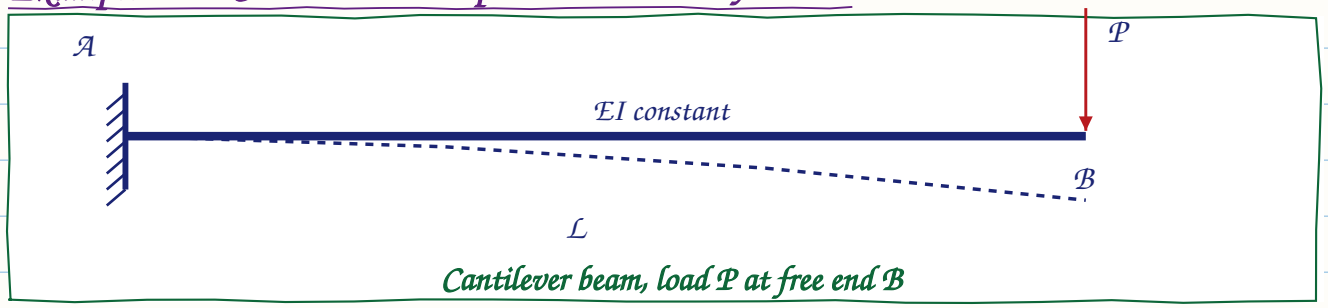
Formula to remember

Beams: $\delta = \int (M/EI)(\partial M/\partial P)\, dx$

Trusses: $\delta = \Sigma (FL/AE)(\partial F/\partial P)$

8. Deflection of Simple Beams — Worked Examples

Example 1 — Cantilever with point load at the free end



Cantilever beam, load P at free end B

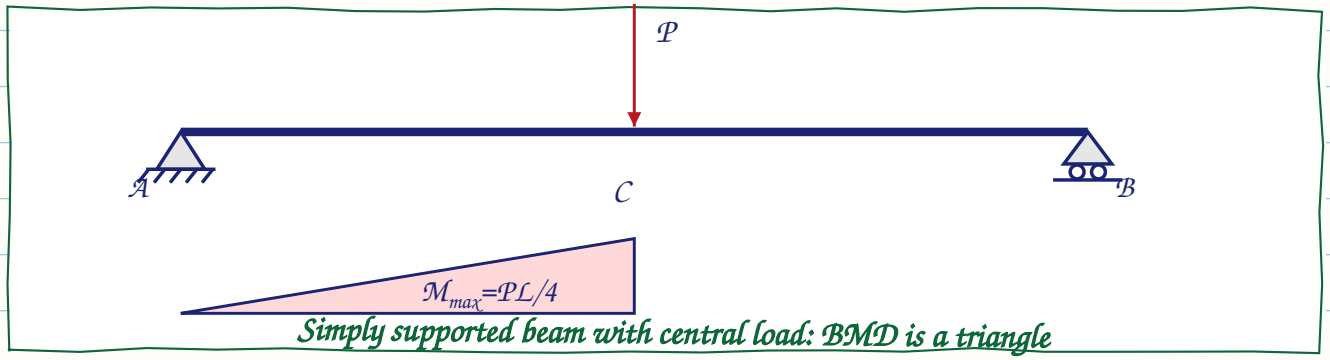
- Take x from the free end. $M = -P\,x$ (hogging); $\partial M/\partial P = -x$.
- $U = \int_0^L P^2x^2\,dx / 2EI = P^2L^3/6EI$.
- $\delta_B = \partial U/\partial P = P\,L^3/3EI$ (downwards).
- Using dummy moment M_0 at B: $M = -Px - M_0$, $\partial M/\partial M_0 = -1$; $\theta_B = \int (Px)(1)/EI\,dx = P\,L^2/2EI$.

Example 2 — Cantilever with UDL w over full span (deflection at free end)

- Free end has no point load, so add a **dummy load P** at B.
- $M = -(P\,x + w\,x^2/2)$; $\partial M/\partial P = -x$.
- $\delta_B = (1/EI)\int_0^L (P\,x + w\,x^2/2)\,x\,dx$, put $P = 0 \rightarrow (1/EI)\int w\,x^3/2\,dx = w\,L^4/8EI$.

Example 3 — Simply supported beam, central point load P

- Reactions $P/2$ each. For $0 \le x \le L/2$: $M = P\,x/2$; $\partial M/\partial P = x/2$.
- Symmetric, so double the first half: $\delta_C = (2/EI)\int_0^{L/2} (Px/2)(x/2)\,dx = (P/2EI) \times (L/2)^3/3 = P\,L^3/48EI$.



Simply supported beam with central load: BMD is a triangle

- $M_{max} = P\,L/4$ at C.

Example 4 — Simply supported beam with UDL (central deflection)

- Add a dummy load P at midspan C. Reaction $R_A = wL/2 + P/2$.
- $0 \le x \le L/2$: $M = (wL/2 + P/2)\,x - w\,x^2/2$; $\partial M/\partial P = x/2$.
- $\delta_C = (2/EI)\int_0^{L/2} (\,wLx/2 - w\,x^2/2\,)(\,x/2\,)\,dx$ $(P = 0)$
- $= (1/EI)\,[\,wL\,x^3/6 - w\,x^4/8\,]_0^{L/2} = (w/EI)(\,L^4/48 - L^4/128\,)$

Result

$\delta_C = 5\,w\,L^4 / 384\,EI$ (downward)

Example 5 — Numerical

Problem

A cantilever of span 3 m carries a point load 20 kN at the free end. $EI = 5 \times 10^4$ kN·m². Find the deflection and slope at the free end.

- $\delta = P\,L^3 / 3EI = 20 \times 27 / (3 \times 50000) = 540 / 150000 = 0.0036$ m = **3.6 mm** .
- $\theta = P\,L^2 / 2EI = 20 \times 9 / (2 \times 50000) = 0.0018$ rad = **0.103°** .

9. Standard Results (Remember These)

Beam and loading	Max deflection δ	Max slope θ
Cantilever, point load P at free end	$P\,L^3 / 3EI$	$P\,L^2 / 2EI$
Cantilever, UDL w over full span	$w\,L^4 / 8EI$	$w\,L^3 / 6EI$
Cantilever, couple M_0 at free end	$M_0\,L^2 / 2EI$	$M_0\,L / EI$
S.S. beam, central point load P	$P\,L^3 / 48EI$	$P\,L^2 / 16EI$
S.S. beam, UDL w over full span	$5\,w\,L^4 / 384EI$	$w\,L^3 / 24EI$
S.S. beam, couple M_0 at one end	$M_0\,L^2 / 9\sqrt{3}\,EI$	$\theta_A = M_0\,L / 3EI$, $\theta_B = M_0\,L / 6EI$
S.S. beam, point load P at distance a from A ($b = L - a$)	under the load: $P\,a^2\,b^2 / 3EI\,L$	$\theta_A = P\,a\,b\,(L + b) / 6EI\,L$

10. Deflection of Pin-Jointed Trusses

- In a pin-jointed truss each member carries only **axial force F** . So strain energy:

Truss strain energy

$U = \Sigma\, F^2\,L / 2AE$ (sum over all members)

- Castigliano: $\delta = \partial U / \partial P = \Sigma\, F\,(\partial F / \partial P)\,L / AE$.
- $\partial F / \partial P = k$ = force in the member due to a **unit load** at the joint and in the direction of the required deflection (where P is the load applied there, real or dummy).

Working formula for trusses

$\delta = \Sigma\, F\,k\,L / AE$ (F = force due to actual loads, k = force due to unit load)

- Tension +ve, compression –ve (the product $F\,k$ is positive when both have the same sign).
- If the load acts at the joint, use that load P itself as the variable: $F = F_0 + kP$ form.

Procedure

Find member forces F (method of joints) → Remove loads, apply unit load at required joint and direction → Find k in each member → Tabulate F, k, L, A, E

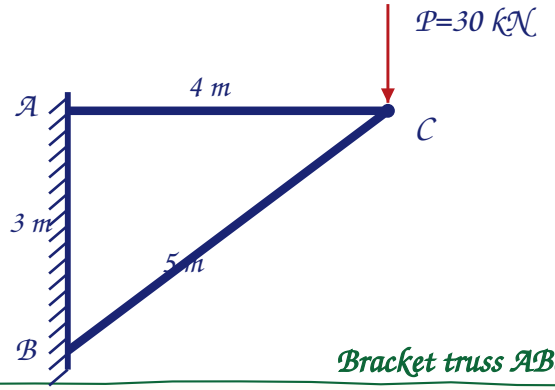
$\delta = \Sigma F k L / AE$

- For a **horizontal deflection** apply a horizontal unit (dummy) load; for vertical, a vertical unit load.
- Zero-force members ($F = 0$ and $k = 0$) contribute nothing, but may carry k even if $F = 0$.

Example 6 — Bracket truss

Problem

Two-member truss: AC horizontal (4 m), BC inclined (5 m), A and B on a wall 3 m apart. A vertical load $P = 30$ kN acts at C. $AE = 1 \times 10^5$ kN for both members. Find the vertical deflection of C.



Bracket truss ABC with load P at C

- Joint C: $F_{BC} \sin\theta = P$ with $\sin\theta = 3/5 \rightarrow F_{BC} = 50$ kN (compression); $F_{AC} = F_{BC} \cos\theta = 40$ kN (tension).
- Unit vertical load at C: $k_{BC} = -5/3$, $k_{AC} = +4/3$.

Member	F (kN)	k	L (m)	FkL
AC	+40	+4/3	4	213.33
BC	−50	−5/3	5	416.67
Total				630

- $\delta_C = \Sigma F k L / AE = 630 / (1 \times 10^5) = 0.0063$ m = **6.3 mm** (downward).

11. Dummy Load Method for Trusses — Points

- Used for: vertical deflection of the bottom chord joints, horizontal movement of a roller joint, relative displacement between two joints (apply equal and opposite unit loads along the line joining them).

- **Lack of fit / temperature:** extra term $\Sigma\, k(\lambda)$ where λ = member elongation due to temperature ($\alpha\, T\, L$) or lack of fit; $\delta = \Sigma\, F\, k\, L/AE + \Sigma\, k\lambda$.
- Check: if loads are applied at several joints, $F = F_0 + P\, k$ form; $\partial F/\partial P$ with P the particular load.

→ A common exam mistake: forgetting the sign of F and k. Always write tension as + and compression as −.

11A. More Worked Examples on Beams

Example 7 — S.S. beam with an eccentric point load



S.S. beam, point load P at distance a from A ($b = L - a$)

- Reactions: $R_A = P\, b / L$, $R_B = P\, a / L$.
- Segment AC ($0 \le x \le a$ from A): $M = P\, b\, x / L$. Segment BC ($0 \le x' \le b$ from B): $M = P\, a\, x' / L$.
- $U = (1/2EI)\,[\int_0^a (P\, b\, x/L)^2\, dx + \int_0^b (P\, a\, x'/L)^2\, dx'] = P^2 a^2 b^2 (a + b) / 6EI\, L^2 =$ **$P^2 a^2 b^2 / 6EI\, L$** .
- Deflection under the load: $\delta_C = \partial U/\partial P =$ **$P\, a^2 b^2 / 3EI\, L$** .
- Check: for $a = b = L/2 \rightarrow P\, L^3 / 48EI$ √ (Example 3).

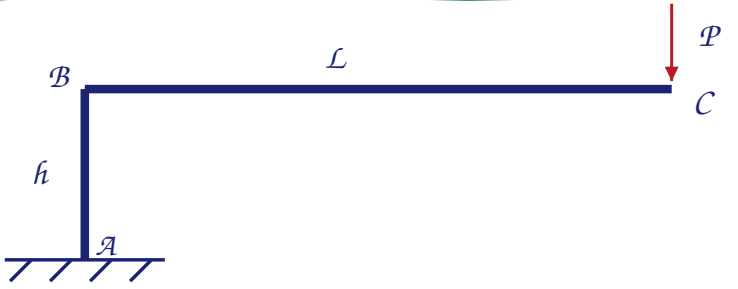
Example 8 — Strain energy of an S.S. beam with UDL

- $M = wLx/2 - wx^2/2 \rightarrow \int_0^L M^2\, dx = w^2 L^5 / 120$.
- **$U = w^2 L^5 / 240\, EI$** .

Example 9 — Cantilever carrying UDL and a point load at the free end

- $M = -(P\, x + w\, x^2/2)$. Using the superposition of the **deflections** (not of energies):
- **$\delta_B = P\, L^3 / 3EI + w\, L^4 / 8EI$** .

Example 10 — L-shaped bracket (frame) — vertical deflection of C



Bracket ABC fixed at A, load P at C

- Member CB (x from C): $M = -P\,x$; $\partial M/\partial P = -x$. Member BA (y from B): $M = -P\,L$ (constant); $\partial M/\partial P = -L$.
- $\delta_C = (1/EI)\,[\int_0^L P\,x^2\,dx + \int_0^h P\,L^2\,dy\,] = P\,L^3/3EI + P\,L^2\,h\,/\,EI$.

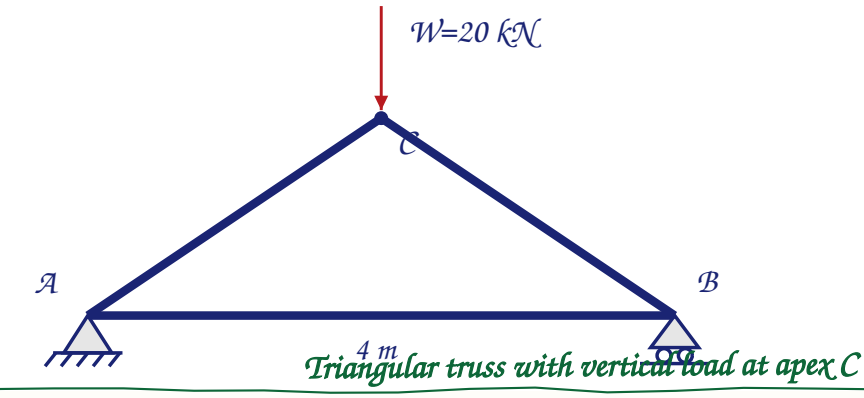
Result

$\delta_C = P\,L^2\,(L + 3h)\,/\,3EI$ (axial energy of AB neglected)

- Same EI is assumed for both members. If axial effect in AB is included add $P\,h\,/\,AE$.

11B. More Worked Examples on Trusses

Example 11 — Triangular truss (span 4 m, rise 3 m, load 20 kN at apex)



Triangular truss with vertical load at apex C

- $R_A = R_B = 10$ kN. Length $AC = BC = \sqrt{(2^2 + 3^2)} = \sqrt{13} = 3.606$ m.
- Joint A: $F_{AC}\sin\varphi = 10$ with $\sin\varphi = 3/\sqrt{13} \rightarrow$ **$F_{AC} = F_{BC} = -12.02$ kN** (compression); **$F_{AB} = +6.67$ kN** (tension).
- Unit vertical load at C: $k_{AC} = k_{BC} = -0.601$, $k_{AB} = +0.333$.

Member	F (kN)	k	L (m)	F k L
AC	−12.02	−0.601	3.606	26.05
BC	−12.02	−0.601	3.606	26.05
AB	+6.67	+0.333	4.00	8.89
Total				60.99

- With $A = 1000$ mm² and $E = 200$ GPa: $AE = 2 \times 10^5$ kN $\rightarrow$ **$\delta_C = 60.99 / 2\times10^5 = 3.05 \times 10^{-4}$ m ≈ 0.305 mm** (downward).
- Horizontal movement of roller B: apply a horizontal unit load at B: only AB gets $k = +1$; $\delta_B =$

$(6.67)(1)(4) / 2\times10^5 = 0.133$ mm.

11C. Strain Energy — Numerical and Extra Concepts

Example 12 — Stepped bar

Problem

A steel bar 2 m long: first 1 m has $A = 400$ mm², next 1 m has $A = 200$ mm². $P = 40$ kN axial; $E = 200$ GPa. Find strain energy and elongation.

- $U = (P^2 / 2E)(L_1/A_1 + L_2/A_2) = [(40\times10^3)^2 / (2 \times 200\times10^9)] \times (1/400\times10^{-6} + 1/200\times10^{-6})$
- $= 0.004 \times 7500 =$ **30 N·m (J)**.
- Elongation: since $U = \frac{1}{2} P \delta \rightarrow \delta = 2U/P = 60/40000 =$ **1.5 mm**.

Shear energy is small — comparison for a cantilever (rectangular section b × d)

- $U_{bending} = P^2 L^3 / 6EI$; $U_{shear} = \alpha P^2 L / 2GA = 3 P^2 L / 5GA$ ($\alpha = 6/5$).
- Ratio $U_{shear} / U_{bending} = (3/10)(E/G)(d/L)^2 \approx 0.78 (d/L)^2$ (taking $E/G = 2.6$).
- For $d/L = 1/10$, shear energy is only **0.78%** of bending energy → neglected for slender beams. It matters for deep beams ($d/L \geq 1/4$).

Castigliano's theorem in its force form

- If U is expressed in terms of **displacements** Δ: $P = \partial U / \partial \Delta$ (force corresponding to a displacement). Example: bar $U = \frac{1}{2} (AE/L)\Delta^2 \rightarrow P = AE\Delta / L$ √. This is the basis of **stiffness (displacement) methods** like slope-deflection (Unit IV).

Advantages and limitations of energy methods

Aspect	Advantages	Limitations
Scope	works for beams, trusses, frames, arches	only for linear elastic behaviour
Calculation	no need of the deflected shape	integration may be long for complex M
Deflection at any point	by using a dummy load	extra analysis with the dummy load
Indeterminate analysis	leads to least-work method (Unit II)	supports must be unyielding

11D. Comparison of Methods of Finding Deflection

Method	Basis	Best suited for
Double integration	$EI\, d^2y/dx^2 = M$	simple beams, single loading
Macaulay's method	integration with singularity terms	beams with many loads
Moment-area method	area of M/EI diagram	slope and deflection at a few points

Method	Basis	Best suited for
Conjugate beam method	M/EI as load on a conjugate beam	prismatic or varying EI beams
Castigliano / unit load	strain energy, partial derivative	trusses, frames, any point
Virtual work	external virtual work = internal	general structures

11E. Common Mistakes and Exam Tips

- Differentiate **before** integrating — saves effort and errors.
- A dummy load P must be **set to zero** after differentiation.
- Use correct **limits** for each segment; the origin of x may differ per segment.
- **Sign** of M need not be consistent across segments, but M and $\partial M/\partial P$ must follow the same sign convention in the same segment.
- State the **direction** of the deflection (downward/upward) in the final answer.
- In trusses write a **table** (Member, L, F, k, F k L) and total it — easy marks.
- Write the **units** : EI in kN·m², AE in kN, results in mm.

→ Quick formula: cantilever $\delta = PL^3/3EI$, UDL cantilever = $wL^4/8EI$, S.S. centre P = $PL^3/48EI$, S.S. UDL = $5wL^4/384EI$ — these may be used to check your Castigliano results.

12. Maxwell's Reciprocal Theorem and Betti's Law (supporting)

- **Betti's law:** In a linear elastic body, work done by a first system of forces on the displacements caused by a second system = work done by the second system on the displacements caused by the first. $P_1\delta_{12} = P_2\delta_{21}$ (δ_{12} = deflection at 1 due to load at 2).
- **Maxwell's reciprocal theorem:** The deflection at point A due to a unit load at B equals the deflection at B due to a unit load at A. $\delta_{AB} = \delta_{BA}$ (also for rotations and mixed).
- Application: influence lines by Müller–Breslau, and symmetry of flexibility and stiffness matrices.

13. Principle of Virtual Work / Unit Load Method (link)

- Unit load method gives the same result as Castigliano: $\delta = \int M\,m\,dx/EI$ where m = BM due to a unit load at the point and in the direction of δ.
- **Castigliano** $(\partial M/\partial P) = m$ when the load is varied: so both methods coincide. Castigliano's advantage: for loads that already act at the point, no new analysis needed.

14. Quick Revision — Unit I

Standard deflections

$U_{axial} = P^2 L / 2AE$ | $U_{bending} = \int M^2 dx / 2EI$ | $U_{shear} = \int \alpha V^2 dx / 2GA$ | $U_{torsion} = T^2 L / 2GJ$ | Castigliano: $\delta = \partial U/\partial P$, $\theta = \partial U/\partial M$ | Beam: $\delta = \int (M/EI)(\partial M/\partial P)\, dx$ | Truss: $\delta = \Sigma F k L/AE$.

Cantilever: $PL^3/3EI$, $wL^4/8EI$ | S.S.: $PL^3/48EI$, $5wL^4/384EI$.

Possible exam questions

1. Derive the expressions for strain energy due to axial load, bending moment and shear force.
2. State and prove Castigliano's first theorem.
3. Find the deflection at the free end of a cantilever with a UDL using Castigliano's theorem.
4. Determine the central deflection of a simply supported beam carrying a UDL.
5. Find the vertical deflection of a joint of the pin-jointed truss given (numerical).
6. State Maxwell's reciprocal theorem.